

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - VIII

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BeautifulSoup

1.Extract Headings

Write a Python program using BeautifulSoup to extract and print all < h2 > headings from a webpage.

```
import requests
from bs4 import BeautifulSoup
```

```
url = "https://www.google.com"

response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

# Get the page title
print(soup.title.text)

# Get all headings
for heading in soup.find_all("h2"):
    print(heading.text.strip())
```

Google

2.Extract Links

Write a Python program using BeautifulSoup to extract all < a > tags and print their link text and URLs.

```
url = "https://www.google.com"

response = requests.get(url)

soup = BeautifulSoup(response.text, "html.parser")

links = soup.find_all("a")

for link in links:
    text = link.get_text(strip=True)
    href = link.get("href")

    print("Link Text:", text)
    print("URL:", href)
    print("-" * 40)
```

```
Link Text: Gmail
URL: https://mail.google.com/mail/&ogbl
-----
Link Text: Images
URL: https://www.google.com/imghp?hl=en&ogbl
-----
Link Text:
URL: https://www.google.com/intl/en/about/products
-----
Link Text: Sign in
URL: https://accounts.google.com/ServiceLogin?hl=en&passive=true&continue=https://www.google.com/&ec=GAZAmgQ
-----
Link Text: Advanced search
URL: /advanced_search?hl=en&authuser=0
```

```
-----  
Link Text: Advertising  
URL: /intl/en/ads/  
-----  
Link Text: Business Solutions  
URL: /services/  
-----  
Link Text: About Google  
URL: /intl/en/about.html  
-----  
Link Text: Privacy  
URL: /intl/en/policies/privacy/  
-----  
Link Text: Terms  
URL: /intl/en/policies/terms/  
-----
```

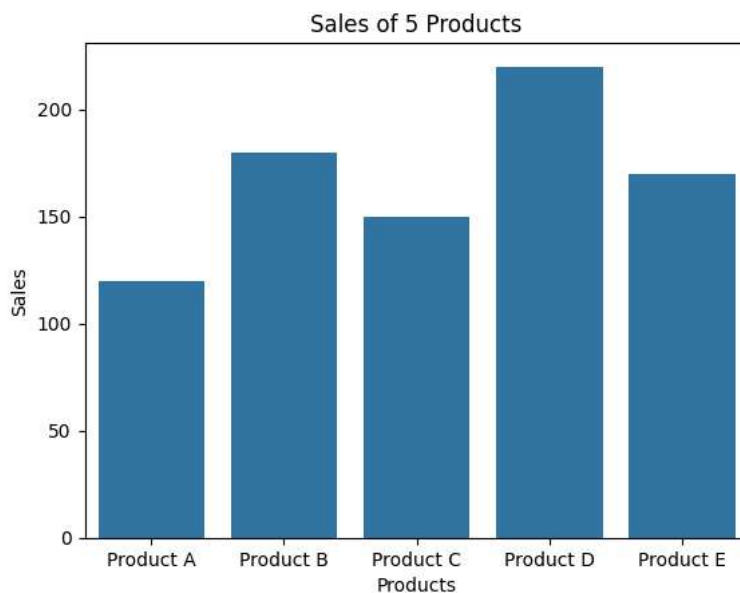
Seaborn

```
import matplotlib.pyplot as plt  
import seaborn as sns
```

3.Bar Chart

Create a Seaborn bar chart showing the sales of 5 different products.

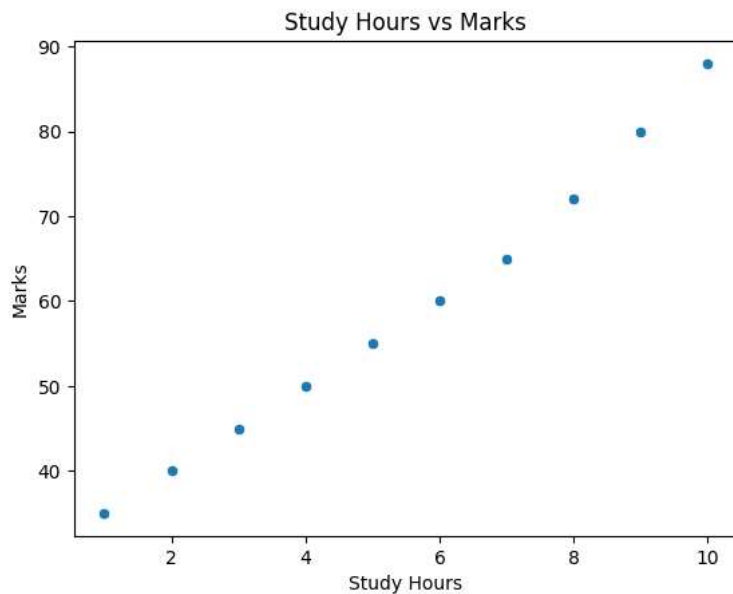
```
products = ["Product A", "Product B", "Product C", "Product D", "Product E"]  
sales = [120, 180, 150, 220, 170]  
  
sns.barplot(x=products, y=sales)  
  
plt.title("Sales of 5 Products")  
plt.xlabel("Products")  
plt.ylabel("Sales")  
plt.show()
```



4.Scatter Plot

Create a Seaborn scatter plot showing the relationship between students' study hours and their marks.

```
study_hours = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
marks = [35, 40, 45, 50, 55, 60, 65, 72, 80, 88]  
  
sns.scatterplot(x=study_hours, y=marks)  
  
plt.title("Study Hours vs Marks")  
plt.xlabel("Study Hours")  
plt.ylabel("Marks")  
plt.show()
```



5.Histogram

Create a Seaborn histogram to display the distribution of students' marks.

```
marks = [45, 67, 78, 56, 89, 92, 73, 65, 81, 55,  
         48, 70, 76, 84, 91, 60, 68, 72, 88, 95]  
  
sns.histplot(marks )  
  
plt.title("Distribution of Students' Marks")  
plt.xlabel("Marks")  
plt.ylabel("Number of Students")  
plt.show()
```

